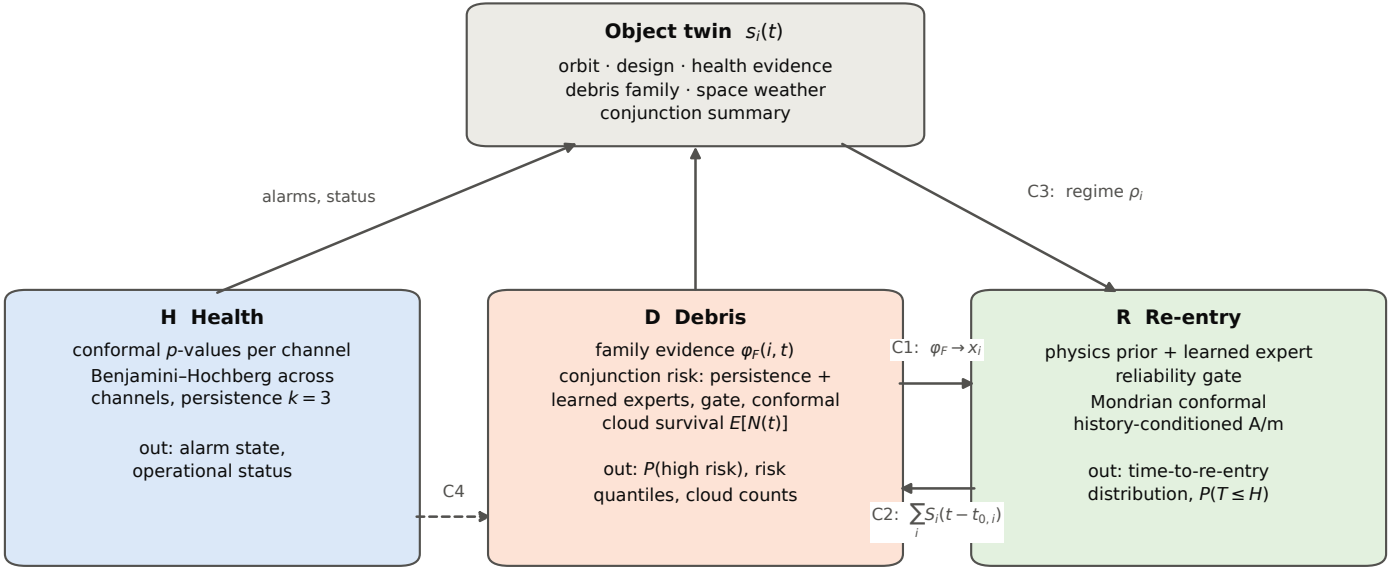

coupling	direction	variable that crosses the boundary	evaluation
C1	debris → re-entry	breakup- and launch-family statistics known at t_0 enter the expert features	ablation, 2 origins
C2	re-entry → debris	per-fragment survival summed over a breakup cloud gives its expected population	61 clouds
C3	health → re-entry	operational status selects the decay regime; drag forecast withheld when controlled	13,794 payloads
C4	health × debris	alarm or non-operational status escalates a high-risk conjunction	no public paired data

Solid arrows: couplings tested on real data. Dashed: a demonstration, because no public dataset pairs telemetry with conjunction messages for the same spacecraft.